Title : Spatial Riemann Zeta Fun: A Scenic Prime Climb

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Spatial Riemann Zeta Fun

A Scenic Prime Climb

$\zeta(s) = \Sigma\ n\text{–}s$

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PrefaceThe original aim of this book was to finally express Riemann's extraordinary spatial intuition—particularly his enhancement of the Zeta function—in the "inexhaustively fruitful" language appropriate to it; namely the language that subjects "points, lines and so forth directly to calculation." That language is Grassmann's, and those quotes are from his book introducing it.

Unfortunately, this book has not quite reached the foothills of the prime climb. My best-used-by date has passed; and I am flinging it out now in the hope that a half-century rassling with various geometric languages might benefit neophytes and sophisticates in different ways, like so:

Up to the Unifying Clifford's Unification chapter, neophytes may get a fresh perspective on the basics of geometric algebra. Then, entering that chapter, sophisticates might encounter a fresh perspective on Clifford Algebra. Everyone, I am hoping, will find delight in Grassmann's language that subjects geometric numbers directly to calculation.

He, like Riemann, had a rich spatial intellect; and he too invested much time visualizing mathematics. Moreover, he was a contemporary of Riemann, a not-so-distant neighbor in fact, a German-speaking compatriot.

Unlike Riemann however, Grassmann was an unknown amateur; a schoolteacher, father of eleven children, nine who survived when survival was not easy. His personal struggle to survive prompted a request at the end of the Foreword in his first book for an "indulgence, that the time for my research was extremely paltry".

By contrast, abundant time for research was necessary for Riemann's own struggle to survive—he began preparing a path thru life by enrolling as a student of the most renowned mathematician in the world at that time: Carl Friedrich Gauss.

For such a path, and with such a teacher, unending research was expected; and crucial to even graduate to the starting point. Upon arrival there, Riemann inherited the circumspect audience of his teacher: the most luminary mathematicians of that era.

That audience, Grassmann declared, was the "object of my striving", but his persistent obscurity denied it to him. And that obscurity, unbeknownst to Riemann, denied him a superior language for engaging that audience.

But there was a brief moment in history—had its outcome been different—that would have given both men what they were denied. It would have also given us what we have been denied: our own ability to subject points, lines and so forth directly to calculation.

Such extraordinary expressiveness could have been acquired easily in youth by most of us, as spatial arithmetic, a universal kind of mathematical mother tongue. Now instead, only a few of us struggle with it as adults, as an esoteric foreign language.

The brief moment that denied us that language hinges on Gauss's contrary responses to Grassmann and Riemann. Such contrariness likely could not have been otherwise—Gauss's immense renown had effectively baked it into his personality:

He came to hate responding to mathematical enquiries because he had typically already resolved them himself in happy silence. He also came to hate teaching because it intruded upon that silence; and he found in his students little of the immense potential he himself had exhibited as a student.

The lone exception was student Riemann. His habilitation lecture, “On the Hypotheses that Lie at the Foundations of Geometry”, inspired this comment from Gauss: “A substantial and valuable work, which does not merely meet the standards required for a doctoral dissertation, but far exceeds them.” Coming from him, such praise was truly exceptional, astonishingly exceptional.

Unexceptional however, was Gauss’s response to the unknown Grassmann, who sent him (and every other mathematical luminary whose address he could find) a copy of his Linear Extension Theory, proudly subtitled A New Branch of Mathematics.

Gauss successfully penetrated all the way to the end of its Foreword where he encountered an idea that provoked an abrupt reply, dated December 14, 1844:   Gauss stated he had worked on similar ideas himself a half century before, and published some results in 1831. (See A History of Vector Analysis, Crowe, p78)

He continued that he was very busy, and that the only way to get at the kernel of Grassmann’s work was to familiarize himself with its “peculiar terminology”. We may interpret this as Gauss’s polite way of saying that he would never read Grassmann’s book.

(Doing so would have been out of character—Gauss had stopped reading math books early in his life, when he began to write them instead. He did try to read the new book by his ex-student, August Mobius, entitled Barycentric Calculus. It developed an addition of points before Grassmann had, but unknown to him (and much less expressive—it did not include a multiplication of points—it was not an arithmetic of points, as Grassmann’s algebra was).

Gauss found Mobius’s book nearly as impenetrable as Grassmann’s, despite its conventional terminology and Cartesian concepts. He did give it the faint praise due, at least, for the earnest effort of a former student; one who had not quite followed in the Master's footsteps, as Riemann had.

¿What at the end of Grassmann's Foreword provoked Gauss's response? It was the idea that an area, "with the plane of rotation retained" can represent an imaginary number, "whence all imaginary expressions now acquire a geometric meaning." Grassmann intended to generalize these patches of area to higher spaces in a second volume, presumably to be entitled Angular Extension Theory, "the solution of which I have not had sufficient leisure." Insufficient leisure was the abiding misfortune of his life.

Indeed, he never found enough to pursue imaginaries further because Gauss's condescending response implied that Grassmann's ideas had already been developed by a superior intellect. So Grassmann abandoned his plans for a subsequent Angular Extension Theory; a decision he publicly lamented several times.

Gauss had evidently seized on Grassmann's area idea as a peculiar way of representing complex numbers on a plane, the idea he had published in 1831. This ad-hoc concept had already been published in variant forms by Argand and Wessel decades before (but possibly after Gauss had concocted the idea himself in silence). However, this idea was not at all what Grassmann had described.

Gauss was probably doomed to misinterpret it that way because he—like his entire audience of luminaries—had long considered an "imaginary" to be an ordinary number needlessly obscured by that unfortunate Cartesian term. He wrote:

"That this subject has hitherto been surrounded by mysterious obscurity, is to be attributed largely to an ill-adapted notation. If, for example, +1, –1, and the square root of –1 had been called direct, inverse and lateral units, instead of positive, negative and imaginary (or even impossible), such an obscurity would have been out of the question."

Gauss's terms "direct, inverse and lateral units" clearly envision his complex plane, in which imaginary numbers do indeed proceed laterally—sideways—to "direct" numbers, which proceed forward, and "inverse" numbers, which proceed backward. This quote became viral in our mathematical community, and has essentially locked us into two centuries of having i considered an ordinary number like 1, 2 … e, π …

This need not have happened; and I have a two-part fantasy about that. The first part deals with Grassmann's obscure conception about imaginaries just described; little appreciated when he described it, and still largely unknown even now. To make it known, I fantasize that the letter Gauss wrote to Grasmann on December 14, 1844—that brief moment in history—went like this:

Esteemed Professor Grassmann

When I first breezed over your ideas for a second volume of your book, I thought you were describing the complex plane, an idea I had published in 1831. On closer scrutiny, however, [remember: a fantasy] I realized your idea is much deeper and far more expressive than mine, altho its algebraic visualization actually generates my idea.

You are describing a unit planar number that can multiply a line, thereby rotating it by a quarter turn. Moreover this “magnitude”, as you say, has its “plane of rotation retained”, meaning that there are actually endlessly many so-called “imaginaries”, each having a different direction! And you hope to extend such directions to higher spaces in your second volume!

I encourage you to vigorously pursue this—each of us has his own peculiar enthusiasms; but mine now head in a different direction. So I won’t even pretend that I will read your book—I can hardly read anything anymore that I didn’t write myself.

Instead, I will pass it along to my students—they can still read what they haven’t written themselves. (Naturally so—they have hardly written anything yet. Ha.)

Respect and Admiration, Carl

This would have had two wonderful consequences. First, it would have encouraged Grassmann to “vigorously pursue” his ideas, rather than abandoning them. This might have eventually secured for him the luminous audience he strove for. Second, Gauss’s student Riemann would have been exposed to an expressive language for his spatial intuitions.

I suspect Riemann was almost uniquely predisposed to quickly absorb such a language, despite its “peculiar terminology”. The only actual reader of Grassmann to quickly absorb his ideas seems to have been Clifford, largely because he had already developed similar ideas himself. Riemann had also developed similar ideas, uniquely so among his peers.

These two consequences—Riemann's acquisition of Grassmann's language and Grassmann's continued pursuit of spatial imaginaries—might have stemmed the gushing flow about imaginaries, well known to us all, that they are really just ordinary numbers unfairly cloaked in mystery by Descartes' misleading terminology.

Merely stemming a flow of misconception is not ideal however; so my second fantasy is to have had that flow abruptly turned off. The only person with enough authority to have done so is the one who turned it on—Carl Friedrich Gauss. He, I fantasize, had actually written this about imaginary numbers:

That this subject has hitherto been surrounded by mysterious obscurity is to be attributed to a chronic historical misconception: that numbers do not come in various dimensions; an idea that encouraged the perverse idea that a number that squares to minus one must be "imaginary".

However, Professor Hermann Grassmann describes spatial numbers that square to minus one. In particular, he explains that planar-numbers can rotate a lineal-number by a quarter turn. Such multiplication squared is two quarter turns of course, meaning minus one. How "imaginary" is that?

Most astonishing to me, spatial numbers allow us to algebraically visualize ordinary numbers (which are not really points on a horizontal line), and also visualize the so-called "impossible" numbers (which are not really points on a lateral line). This visualization can be done in in a genuine space of point-numbers.

If spatial numbers had been know previously, the obscurity of negative-squaring numbers would have been out of the question. However, these ideas are not yet ripe for publication by me; so I will remain silent for now.

These ideas only became ripe in 1876 (two decades after Gauss had gone silent forever) when William Clifford presented a unification of Grassmann's free sub-algebra to the London Mathematical Society. This unification, when combined with Grassmann's full free-and-bound algebra, allows the precise algebraic visualization just described in my two fantasies.

(If the terms spatial arithmetic, free sub-algebra, free-and-bound algebra *etc.* are unfamiliar, you can make them familiar by mining Felicitous Geometric Algebra and Full Unified Geometric Algebra. I practically give those books away to disseminate the ideas. The ideas in them serve as a foundation for the present book; but they are unconventional and unfamiliar, so they will require some rassling.)

Since this book will be using the just-sketched algebraic visualization thruout, let us quickly fly over it now to prepare for a scenic hike in the next chapter.

The visualization shall deploy spatial arithmetic, but restricted to a plane. This restriction will itself become mostly restricted to the free sub-algebra of the plane. Even that restriction shall ultimately become restricted primarily to the even sub-algebra of the free sub-algebra of the spatial arithmetic of the plane. Such a severe restriction is conventionally known as “the algebra of the complex plane”.

The beauty of all of this is that—on the one hand—it shall expose how relatively simple and transparent the complex plane can be made to be; and—on the other hand—how rich and expressive it can be made to be: just ease the restrictions. You’ll see.

You will literally see.

¿How Shall We Speak?We shall speak in the spatial arithmetic for physical space, but subsequently tailored for Riemann: initially restricted to a plane; then further restricted to its free sub-algebra; finally restricted to its even free sub-algebra. Even so-restricted, this language is very expressive; befitting Riemann’s intricate ideas. It is best approached by stepping up to it from simpler spaces.

This ploy follows the historical path by introducing each new—possibly startling—idea as an enhancement of previous ideas, or a consequence of them. This new language shall be quickly breezed over at first; and finally stepped thru carefully one dimension at a time. By the end of our explorations, startlement should have dissipated; and we can begin marching toward the zeta function.

The simplest actually spatial space is 1-point space, which is obviously a bound space. Let us label its single basis point o because it can be used as the origin in all higher bound spaces. However, it is better termed the bound ceiling in this space because it fills the entire space.

This point, under spatial arithmetic, generates all possible sums and products of itself. This may seem sterile for a single point; but it is surprisingly rich. For starters, sums of a point with itself generate scaled versions of itself.

The scalars so generated can be detached from their points, and then add and multiply with them, thereby generating numbers having composite dimension {0, 1} (where either of these dimensions may be absent like so: {0} or {1}; or possibly both: {}, which is the "dimension" of 0, meaning no dimension at all, not even scalar dimension {0}). These various numbers—and even the non-dimensioned non-number 0—can thence add and multiply themselves. ¿Are you beginning to see how rich even the most rudimentary spatial arithmetic can be?

Its richness owes to cohering and coalescing creativeness absent in ordinary scalar arithmetic. In there, addition and multiplication always first cohere two scalars together and then coalesce them to another scalar, a new creation.

In spatial arithmetic, by contrast, addition and multiplication may or may not coalesce, even tho they always cohere and create. For an obvious example, adding a scalar to a point does not coalesce them to either a scalar or a point. Nonetheless, addition does cohere together these two mono-dimensioned numbers—monos let's call them—to create a new plural-dimensioned number—a plur.

The crucial takeaway is that cohering creativity does not necessarily require, or even imply coalescence, as ordinary arithmetic might lead a person to believe. This idea is crucial because it encompasses even summands having the same dimension: they may nonetheless be so different that they fail to coalesce. This idea may be the most startling one you will encounter in this chapter.

Similarly, multiplication in spatial arithmetic always coheres and creates a new number; but may or may not coalesce it. And happily so: when it does not coalesce to a mono, the plur result is extraordinarily informative as we shall see.

Which introduces the first slight startlement: There are three different kinds of multiplication in spatial arithmetic, but they all arise from a single primitive one. It is not scalar multiplication, as might seem intuitive. That product, in fact, is not even primitive in ordinary arithmetic either.

Ordinary arithmetic begins with one primitive operation: addition. This naturally spawns multi-addition, unsurprisingly, meaning scalar multiplication.

Similarly, the most primitive operation in spatial arithmetic is also addition, which also naturally spawns multi-addition. This more general spatial kind of scalar multiplication is thence augmented by the most primitive spatial product, namely extension. It vanishes for a point extended with itself; but it is introduced now because such vanishing has important ceiling consequences in higher spaces.

This product does not derive from anything more fundamental—it is an afore-mentioned enhancement that Grassmann introduced in the early 1800's. It naturally spawns two more products, perhaps surprisingly. Least surprising may be its own undoing, namely retraction.

Historically, undoing operations like this have been a font of creativity: subtraction, as an pre-history example; and division, an early-history one, are quite familiar. The just-mentioned undoing—not likely familiar—can be combined with its doing to create extension-retraction. Three spatial products altogether, a close-knit doing-undoing family of them.

Which brings up a second not-so-slight startlement: when spatial extension-retraction has a non-spatial multiplicand (a scalar) it becomes multi-addition. In other words, in spatial arithmetic scalar multiplication is a vacuous kind of extension-retraction.

This means that scalar multiplication in that language is ultimately spawned both by the primitive sum, addition; and also by the primitive spatial product, extension. The historical path to this enlightenment was long and winding, so let us just jog thru it for now:

Spatial-product meanderingsGrassmann used only extension as the spatial product in his first book, augmented by ad-hoc scalar multiplication. He had another name for extension: the outer product because its engagement with a free vector pays attention only to the vector's outer part, its rejection—its perpendicular part.

That made him think there must be a product that pays attention to the vector's inner part, its projection—its parallel part. He called this the inner product in the Foreword to his first book, even tho it never appeared in that book.

That surprising lacuna indicates that he had been thinking about it inconclusively—it was clearly one of those ideas for which he "had not had sufficient leisure".

However, immediately after publishing this book, a prize was offered for implementing Leibniz's Universal Algebra of Everything. Grassmann submitted a very hastily composed inner product as the shining exemplar of that algebra. It was the only submission; and the prize committee asked August Mobius to evaluate it.

Mobius clearly appreciated Grassmann's ingenuity; but commented that it "lies rather far from standard mathematical practice." Grassmann evidently realized that too; and abandoned this attempt soon after it had won the prize. He then began developing an inner product more leisurely via articles in Crelle's Journal.

Finally, in 1862, he presented a coherent inner product in his last book, Extension Theory. In there he belatedly recognized it to be an undoing of extension (but he never used the obvious antonym retraction). He still augmented both of these products with ad-hoc scalar multiplication.

The need for such augmentation unexpectedly disappeared when Clifford unified Grassmann's two products into a single one, which pays attention to both the rejection and projection of a free vector.

Clifford had realized that Hamilton’s meandering syntactic path to the quaternions had generated a Scalar Part that was essentially an inner product with a negated sign; and a Vector Part that was essentially an outer product unwittingly reflected from physical space. Hamilton’s peculiar path had fortuitously combined these two operations.

Such combining is very informative, a fact appreciated by Clifford even tho others like Heaviside and Gibbs deplored it. Clifford proposed in 1876 to “extend this assumption to the Grassmann representation.”

He did so in a simple but startling way: “I take n units i1 i2…in, such that is2 = +1, and ir is = –is ir”. This is startling because the juxtaposed units on the right neg-commute, implying that the squared unit on the left should be 0 rather that +1, as Clifford declared. Some very able mathematicians were initially baffled that Clifford could have made such a sophomoric blunder.

Well he didn’t—he was tacitly using orthonormal free basis vectors whose inner product vanishes when the outer product does not, and vise-versa. Moreover, these two products generate different dimensions—{0} and {2}—which can therefor be added together without interfering with each other, and without losing information.

This is our first example of a product that coheres and creates, but does not necessarily coalesce. Its lack of coalescence, however, only manifests itself on a plane or higher space because only there can a free vector have simultaneous rejection and projection. Such dual information shall become crucial in later explorations.

In other words, Clifford’s new product was an extension plus a retraction: extraction let’s call it hereafter for pithy brevity. His basis-oriented bottom-up approach to it had an unexpected benefit: when his basis products are augmented to include scalars, they reduce to simple scalar multiplication.

Clifford did not name this product, but Hestenes termed it the geometic product; a name used in most exposition on this subject. I prefer products to have pithy verb names; so I shall use extract and extraction in harmony with extend and extension, retract and retraction.

Extraction, rather than extension, is typically considered the primitive product in conventional geometric algebra. Extension is then derived from that; as also is retraction.